

CORPORATE FINANCE – BERK AND DEMARZO GLOBAL EDITION 4TH

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CHAPTER 4: THE TIME VALUE OF MONEY

4.1 THE TIMELINE

Stream of cash flows

A series of cash flows lasting several periods

We can represent them on a timeline.

Timeline

A linear representation of the timing of the expected cash flows.

4.3 THE THREE RULES OF TIMETRAVEL

Rule 1: Comparing and combining values

It's only possible to compare or combine values at the same point in time.

Rule 2: Moving cash flows forward in time

To move a cash flow forward in time, you must compound it.

Compounding: Process of moving a value or cash flow forward in time. Multiply by the interest rate factor $(1+r)$ as needed.

Time value of money: the difference in value between money today and money in the future.

Compound interest: earning interest on interest.

$$FV_n = C * (1+r)^n$$

Rule 4: Moving cash flows back in time

To move a cash flow back in time, you must discount it.

Discounting: Moving a value or cash flow backward in time. We divide it by the interest rate factor.

$$PV = \frac{C}{(1+r)^n}$$

4.3 VALUING A STREAM OF CASH FLOWS

$$PV = \sum_{n=0}^N \frac{C_n}{(1+r)^n}$$

4.4 CALCULATING THE NET PRESENT VALUE

$$NPV = PV(\text{benefits}) - PV(\text{costs}) = PV(\text{benefits} - \text{costs})$$

Benefits are cash inflows and the costs are the cash outflows.

4.5 PERPETUITIES AND ANNUITIES

Perpetuity

A stream of equal cash flows that occur at regular intervals and last forever.

$$PV(C \in \text{perpetuity}) = \frac{C}{r}$$

The PV formula for the perpetuity already discounts the cash flows to one period prior to the first cash flow.

Annuity

A stream of N equal cash flows paid at regular intervals.

$$PV(\text{annuity of } C \text{ for } N \text{ period with interest rate } r) = \frac{C * 1}{r} * \left(1 - \frac{1}{(1+r)^N}\right) = \sum_{n=1}^N \frac{C}{(1+r)^n}$$

$$FV(\text{annuity}) = PV * (1+r)^N = \frac{C * 1}{r} * ((1+r)^N - 1)$$

Growing perpetuity

A stream of cash flows that occur at regular intervals and grow at a constant rate forever.

$$PV = \sum_{n=1}^{\infty} \frac{C(1+g)^{n-1}}{(1+r)^n} = \frac{C}{r-g}$$

The only viable growing perpetuities are those where the perpetual growth rate is less than the interest rate.

Growing annuity

A stream of N growing cash flows, paid at regular intervals.

$$PV = \frac{C}{r-g} \left(1 - \left(\frac{1+g}{1+r}\right)^N\right) = \frac{C}{r-g}$$

4.6 USING AN ANNUITY SPREADSHEET OR CALCULATOR

Use Excel, basically.

4.7 NON-ANNUAL CASH FLOWS

4.8 SOLVING FOR CASH PAYMENTS

$$C(\text{loan} \vee \text{annuity payment}) = \frac{P}{\frac{1}{r} * \left(1 - \frac{1}{(1+r)^N}\right)}$$

4.9 THE INTERNAL RATE OF RETURN

Internal rate of return (IRR)

The interest rate that sets the net present value of the cash flows equal to zero.

$$IRR \text{ with two cash flows} = \left(\frac{FV}{P}\right)^{\frac{1}{N}} - 1$$

Because we're just comparing two cashflows, the IRR calculation is equivalent to computing the **compound annual growth rate (CAGR)** of the cash flow.

$$IRR \text{ of growing perpetuity} = \frac{C}{P} + g$$

Analyzing the Project

Break-even level of an input

The level for which the investment has an NPV of zero.

Example: IRR, EBIT break-even for sales

Break-even analysis

For each parameter, we calculate the value at which the NPV of the project is zero.

Sensitivity analysis

Breaks the NPV calculation into its component assumptions and shows how the NPV varies as the underlying assumptions change.

Allows us to explore the effects of errors in our NPV estimates for the project.

We learn which assumptions are the most important, and we can then invest further resources and effort to refine these assumptions.

Also reveals which aspects of the project are most critical when we are actually managing the project.

Scenario analysis

Considers the effect on the NPV of changing multiple project parameters.

CHAPTER 7: INVESTMENT DECISION RULES

7.1 NPV AND STAND-ALONE PROJECTS

NPV Investment Rule

When making an investment decision, take the alternative with the highest NPV. Choosing this alternative is equivalent to receiving its NPV in cash today.

➔ Take the highest NPV

NPV Profile

A graph of the project's NPV over a range of discount rates

Internal Rate of Return (IRR)

The discount rate that sets the NPV of the project's cash flows equal to zero.

➔ The difference between the cost of capital and the IRR is the maximum estimation error in the cost of capital that can exist without altering the original decision.

$$\text{Cost of capital} - \text{IRR} = \text{Max} \{ \text{Estimation error} \in \text{the cost of capital} \}$$

7.2 THE INTERNAL RATE OF RETURN RULE

IRR Investment Rule

Take any investment opportunity where the IRR exceeds the opportunity cost of capital.

Turn down any opportunity whose IRR is less than the opportunity cost of capital.

➔ For stand-alones, it only works when all negative cash flows precede the positive ones.

Pitfall #1: Delayed Investments

It's as if you're borrowing money, so you want the lowest rate possible.

Pitfall #2: Multiple IRRs

The IRR rule doesn't apply. We can only rely on the NPV rule.

Pitfall #3: Nonexistent IRR

Literally doesn't exist.

7.3 THE PAYBACK RULE

You should only accept a project if its cash flows pay back its initial investment within a prespecified period.

Payback period

The amount of time it takes to pay back the initial investment.

Payback pitfalls

This rule is typically used for small investment decisions.
Provides budgeting information.

7.4 CHOOSING BETWEEN PROJECTS

Mutually exclusive investments

- **NPV rule:** highest NPV
- **IRR rule:** when projects differ in their scale of investment, the timing of their cash flows, or their riskiness, their IRR cannot be compared.

Incremental IRR

The IRR of the incremental cash flows that would result from replacing one project with the other.

The discount rate at which it becomes profitable to switch from one project to the other.

7.5 PROJECT SELECTION WITH RESOURCE CONSTRAINTS

In principle, the firm should take all projects with positive NPV, but there are often limitations (resource requirements).

Profitability index

Measures the value created in terms of NPV per unit of resource consumed.

$$\text{Profitability index} = \frac{\text{Value created}}{\text{Resource consumed}} = \frac{\text{NPV}}{\text{Resource consumed}}$$

Shortcomings of the PI

Two conditions must be satisfied:

1. The set of projects taken following the profitability index ranking completely exhausts the available resource.
2. There is only a single relevant resource constraint.

CHAPTER 8: FUNDAMENTALS OF CAPITAL BUDGETING

8.1 FORECASTING EARNINGS

Capital Budget

List of the projects and investments that a company plans to undertake during the coming year.

Capital budgeting

Process during which the firms analyze alternative projects and decide which ones to accept.

Ultimate goal is to determine the effect of the decision on the firm's cash flows, and evaluate the NPV of these cash flows to assess the consequences of the decision for the firm's value.

Incremental earnings

The amount by which the firm's earnings are expected to change as a result of the investment decision.

We begin by determining these.

Unlevered net income

Doesn't include any interest expense associated with debt.

Marginal corporate tax rate

The tax rate it will pay on an incremental dollar of pre-tax income.

$$\text{Income tax} = EBIT * \tau_c$$

$$\text{Unlevered net income} = EBIT * (1 - \tau_c) = (\text{Revenues} - \text{Costs} - \text{Depreciation}) * (1 - \tau_c)$$

Opportunity cost

The opportunity cost of using a resource is the value it could have provided in its best alternative use.

Project externalities

Indirect effects of the project that may increase or decrease the profits of other business activities of the firm.

Cannibalization

When sales of a new product displace sales of an existing product.

Sunk cost

Any unrecoverable cost for which the firm is already liable.

Sunk costs have been or will be paid regardless of the decision about whether or not to proceed with the project.

It shouldn't be included in the analysis.

➔ If our decision doesn't affect the cash flow, then the cash flow shouldn't affect our decision.

Overhead expenses

Are associated with activities that are not directly attributable to a single business activity but instead affect many different areas of the corporation.

8.2 DETERMINING FREE CASH FLOW AND NPV

Free cash flow

The incremental effect of a project on the firm's available cash, separate from any financing decisions.

Net working capital (NWC)

The difference between current assets and current liabilities.
The assets that you need to be able to work.

$$\text{Net working capital} = \text{Current assets} - \text{Current liabilities} = \text{Cash} + \text{Inventories} + \text{Receivables} - \text{Payables}$$

Trade credit

The difference between receivables and payables is the net amount of the firm's capital that is consumed as a result of these credit transactions.

Free cash flow

$$\begin{aligned} & \dot{I} (\text{Revenues} - \text{Costs} - \text{Depreciation}) * (1 - \tau_c) + \text{Depreciation} - \text{CapEx} - \Delta \text{NWC} \\ & \dot{I} \text{Unlevered Net Income} + \text{Depreciation} - \text{CapEx} - \Delta \text{NWC} \\ & \dot{I} (\text{Revenues} - \text{Costs}) * (1 - \tau_c) - \text{CapEx} - \Delta \text{NWC} + \tau_c * \text{Depreciation} \end{aligned}$$

Cost of capital for a project

The expected return that investors could earn on their best alternative investment with similar risk and maturity.

$$PV(FCF_t) = \frac{FCF_t}{(1+r)^t} = FCF \frac{F_t * 1}{(1+r)^t}$$

CHAPTER 9: VALUING STOCKS

9.1 THE DIVIDEND-DISCOUNT MODEL

Discounting stock

Because the cash flows are risky, we can't compute their present value using the risk-free rate. Instead, we must discount them based on the **equity cost of capital**.

Equity cost of capital

The expected return of other investments available in the market with equivalent risk to the firms' shares.

$$P_0 = \frac{\dot{I}_1 + P_1}{1 + r_E}$$

Total return

$$r_E = \frac{\dot{I}_1 + P_1}{P_0} - 1 = \frac{\dot{I}_1}{P_0} + \frac{P_1 - P_0}{P_0}$$

Dividend yield Capital Gain
Rate

Dividend yield

The expected annual dividend of the stock divided by its current price.
The percentage return the investor expects to earn from the dividend paid by the stock.

Capital gain

The difference between the expected sale price and purchase price for the stock.

Capital gain rate

The difference between the expected sale price and purchase price for the stock, divided by the current stock price to express the capital gain as a percentage return.

Total return of the stock

The sum of the dividend yield and the capital gain rate.

The expected return that the investor will earn for a one-year investment in the stock.

➔ The expected total return of the stock should equal the expected return of other investments available in the market with equivalent risk

The Dividend-Discout Model

The horizon N is arbitrary

$$P_0 = \frac{\dot{I}_1}{1+r_E} + \frac{\dot{I}_2}{(1+r_E)^2} + \dots + \frac{\dot{I}_N}{(1+r_E)^N} + \frac{P_N}{(1+r_E)^N}$$

Applies to a single N-year investor, who will collect dividends for N years and then sell the stock, or to a series of investors who hold the stock for shorter periods and then resell it.

All investors, with the same beliefs, will attach the same value to the stock, independent of their investment horizons.

9.2 APPLYING THE DIVIDEND-DISCOUNT MODEL**Constant Dividend Growth**

The simplest forecast for the firm's future dividend states that they will grow at a constant rate, g, forever.

$$P_0 = \frac{\dot{I}_1}{r_E - g}$$

The value of the firm depends on the dividend level for the coming year, divided by the equity cost of capital adjusted by the expected growth rate of dividends.

Dividend payout rate

The fraction of its earnings that the firm pays as dividends each year

$$\dot{I}_t = \frac{\text{Earnings}_t}{\text{Shares Outstanding}_t} * \text{Dividend Payout Rate}_t$$

The dividend each year is the firm's earnings per share (EPS) multiplied by its dividend payout rate.

The firm can increase its dividend in three ways:

1. Increase its earnings (net income)
2. Increase its dividend payout rate

3. Decrease its shares outstanding

If all increases in future earning result exclusively from new investment made with retained earnings, then

$$\text{Change in Earnings} = \text{New Investment} * \text{Return on New Investment}$$

Retention rate

The fraction of current earnings that the firm retains

$$\text{New Investment} = \text{Earnings} * \text{Retention Rate}$$

$$\text{Earnings Growth Rate} = \frac{\text{Change in Earnings}}{\text{Earnings}} = \text{Retention Rate} * \text{Return on New Investment}$$

If the firm chooses to keep its dividend payout rate constant, then the growth in dividends will equal growth of earnings.

$$g = \text{Retention Rate} * \text{Return on New Investment}$$

Sustainable growth rate

The rate at which it can grow using only retained earnings

➔ Cutting the firm's dividend to increase investment will raise the stock price if, and only if, the new investments have a positive NPV

Changing Growth Rates

$$P_0 = \frac{\hat{I}_{N+1}}{r_E - g}$$

Dividend discount model with constant long-term growth

$$P_0 = \frac{\hat{I}_1}{1+r_E} + \frac{\hat{I}_2}{(1+r_E)^2} + \dots + \frac{\hat{I}_N}{(1+r_E)^N} + \frac{1}{(1+r_E)^N} \left(\frac{\hat{I}_{N+1}}{r_E - g} \right)$$

Limitations of the Dividend discount model

A tremendous amount of uncertainty is associated with any forecast of a firm's future dividends.

9.3 TOTAL PAYOUT AND FREE CASH FLOW VALUATION MODELS

Share repurchases

The firm uses excess cash to buy back its own stock.

Two consequences for the dividend discount model:

- The more cash the firm uses to repurchase shares, the less it has available to pay dividends.
- By repurchasing shares, the firm decreases its share count, which increases its earnings and dividends on a per share basis.

Total payout model

Values all of the firm's equity, rather than a single share.

We discount the total payouts that the firm makes to shareholders, which is the total amount spend on both dividends and share repurchases. Then, we divide by the current number of shares outstanding to determine the share price.

$$P_0 = \frac{PV(\text{Future total dividends} \wedge \text{repurchases})}{\text{Shares outstanding}_0}$$

- ➔ We discount total dividends and share repurchases and use the growth rate of total earnings (rather than earnings per share) when forecasting the growth of the firm's total payouts.

The Discounted Free Cash Flow Model

Begins by determining the total value of the firm to all investors, both equity and debt

Enterprise Value

The value of the firm's underlying business unencumbered by debt and separate from any cash or marketable securities.

The net cost of acquiring the firm's equity, taking its cash, paying off all debt, and thus owning the unlevered business.

$$\text{Enterprise Value} = \text{Market Value of Equity} + \text{Debt} - \text{Cash}$$

$$\text{Free Cash Flow} = \text{EBIT} * (1 - \tau_c) + \text{Depreciation} - \text{Capital Expenditures} - \text{Increases} \in \text{Net Working Capital}$$

Net Investment

The investment needed to support the firm's growth, above and beyond the level needed to maintain the firm's existing capital.

$$\text{Net Investment} = \text{Capital Expenditures} - \text{Depreciation}$$

Free cash flow

Measures the cash generated by the firm before any payments to debt or equity holders are considered

$$\text{Free Cash Flow} = \text{EBIT} * (1 - \tau_c) - \text{Net Investment} - \text{Increases} \in \text{Net Working Capital}$$

$$V_0 = PV(\text{Future Free Cash Flow of Firm})$$

$$P_0 = \frac{V_0 + \text{Cash}_0 - \text{Debt}_0}{\text{Shares outstanding}_0}$$

- ➔ Difference between the discounted free cash flow model and the dividend discount model is that in the dividend discount mode, the firm's cash and debt are included indirectly through the effect of interest income and expenses on earnings. In the discounted free cash flow model, we ignore interest income and expenses, and then adjust for cash and debt directly

In the free cash flow model, we should use the firm's **weighted average cost of capital (WACC)**

Weighted average cost of capital (WACC)

The average cost of capital the firm must pay to all of its investors, both debt and equity holders.

If the firm has no debt, it's replaced by the equity cost of capital r_E

Reflects the average risk of all of the firm's investments.

$$V_0 = \frac{FCF_1}{1+r_{wacc}} + \frac{FCF_2}{(1+r_{wacc})^2} + \dots + \frac{FCF_N + V_N}{(1+r_{wacc})^N}$$

$$V_N = \frac{FCF_{N+1}}{r_{wacc} - g_{FCF}} = \left(\frac{1+g_{FCF}}{r_{wacc} - g_{FCF}} \right) \cdot FCF_N$$

Present value of	Determines the
Dividend Payments	Stock Price
Total Payouts (All dividends and repurchases)	Equity Value
Free Cash Flow (Cash available to pay all security holders)	Enterprise Value

9.4 VALUATION BASED ON COMPARABLE FIRMS

Method of comparables

We estimate the value of the firm based on the value of other, comparable firms or investments that we expect will generate very similar cash flows in the future.

Valuation multiples

A ratio of the value to some measure of the firm's scale

Price Earnings ratio

Equal to the share price divided by its earnings per share

$$\text{Forward } \frac{P}{E} = \frac{P_0}{EPS_1} = \frac{\text{Dividend Payout Rate}}{r_E - g}$$

Forward P/E

The P/E multiple computed based on its **forward earnings** (expected earnings over the next twelve months)

We can also compute a firm's **trailing P/E** ratio using **trailing earnings**

Enterprise Value Multiples

Represents the total value of the firm's underlying business rather than just the value of equity

$$\frac{V_0}{EBITDA_1} = \frac{FCF_1/EBITDA_1}{r_{wacc} - g_{FCF}}$$

Limitations of Multiples

Doesn't take into account the important differences among firms.

They only provide information regarding the value of the firm relative to the other firms in the comparison set

CHAPTER 11: OPTIMAL PORTFOLIO CHOICE AND THE CAPITAL ASSET PRICING MODEL

11.6 THE EFFICIENT PORTFOLIO AND REQUIRED RETURNS

Portfolio improvement: Beta and the Required Return

If we take an arbitrary portfolio P and try to raise its Sharpe ratio by selling some risk-free asset and investing the proceeds in some investment i , we have two consequences:

1. Expected return: Our expected return will increase by i 's excess return:
 $E[R_i] - r_f$
2. Volatility: We add the risk that i has in common with our portfolio, the rest will be diversified.
 $SD(R_i) * Corr(R_i, R_p)$

Beta of investment i with portfolio P

$$\beta_i^P \equiv \frac{SD(R_i) * Corr(R_i, R_p)}{SD(R_p)}$$

β_i^P measures the sensitivity of the investment i to the fluctuations of the portfolio P.

For each 1% change in the portfolio's return, investment i 's return is expected to change by β_i^P % due to risks that i has in common with P.

$$E[R_i] > r_f + \beta_i^P * (E[R_i] - r_f)$$

Increasing the amount invested in i will increase the Sharpe ratio of portfolio P if its expected return $E[R_i]$ exceeds its required return given portfolio P, defined as:

$$r_i \equiv r_f + \beta_i^P * (E[R_i] - r_f)$$

Required return

The expected return that is necessary to compensate for the risk investment i will contribute to the portfolio.

Is equal to the risk-free investment rate plus the risk premium of the current portfolio P scaled by i 's sensitivity to P, β_i^P

If i 's expected return exceeds this required return, then adding more of it will improve the performance of the portfolio.

Expected returns and the efficient portfolio

As we buy shares of security i , its correlation (and therefore its beta) with our portfolio will increase, ultimately raising its required return until $E[R_i] = r_i$.

At this point, our holdings of security i are optimal.

And vice versa.

If we have no restrictions on our ability to buy or sell securities that are traded in the market, we will continue to trade until the expected return of each security equals its required return, until $E[R_i] = r_i$ holds for all i .

- ➔ A portfolio is efficient if and only if the expected return of every available security equals its required return.

This result implies the following relationship between the expected return of any security and its beta with the efficient portfolio:

$$E[R_i] = r_i \equiv r_f + \beta_i^{eff} * (E[R_{eff}] - r_f)$$

- ➔ The efficient portfolio is the one with the highest Sharpe ratio of any portfolio in the economy
- ➔ We can determine the appropriate risk premium for an investment from its beta with the efficient portfolio

11.7 THE CAPITAL ASSET PRICING MODEL

This model allows us to identify the efficient portfolio of risky assets without having any knowledge of the expected return of each security. Instead, the CAPM uses the optimal choices investors make to identify the efficient portfolio as the market portfolio, the portfolio of all stocks and securities in the market.

The CAPM Assumptions

1. Investors can buy and sell all securities at competitive market prices (without incurring taxes or transactions costs) and can borrow and lend at the risk-free interest rate.
2. Investors hold only efficient portfolios of traded securities – portfolios that yield the maximum expected return for a given level of volatility.
3. Investors have homogeneous expectations regarding the volatilities, correlations, and expected returns of securities.

Supply, demand, and the efficiency of the market portfolio

- ➔ The efficient tangent portfolio of risky securities (the portfolio that all investors hold) must equal the market portfolio

Optimal investing: The Capital Market line

The tangent line that goes through the market portfolio.

All investors should choose a portfolio on the capital market line, by holding some combination of the risk-free security and the market portfolio.

CHAPTER 12: ESTIMATING THE COST OF CAPITAL

12.1 THE EQUITY COST OF CAPITAL

The cost of capital of any investment opportunity equals the expected return of available investments with the same beta

The CAPM Equation for the Cost of Capital (Security Market Line)

$$r_i = r_f + \beta_i * (E[R_{Mkt}] - r_f)$$

Investors will require a risk premium comparable to what they would earn taking the same market risk through an investment in the market portfolio.

12.2 THE MARKET PORTFOLIO

Value weighted portfolio

Equal ownership portfolio

Passive portfolio

Market index

Report the value of a particular portfolio of securities

Price weighted portfolio

Holds an equal number of shares of each stock independent of their size

Index funds

Invest in portfolios made up of market indexes

Exchange Traded Funds (ETF)

Security that trades directly on an exchange, but represents ownership in a portfolio of stocks

Market proxy

A portfolio whose return they believe closely tracks the true market portfolio

12.3 BETA ESTIMATION

A security's beta

Measures the sensitivity of the security's return to those of the market

The appropriate measure of risk for a well-diversified investor

- ➔ Beta corresponds to the slope of the best fitting line in the plot of the security's excess returns versus the market excess return

Linear regression

The statistical technique that identifies the best fitting line through a set of points

Alpha α_i

Measures the historical performance of the security relative to the expected return predicted by the security market line

12.4 THE DEBT COST OF CAPITAL

The cost of capital that a firm must pay on its debt

12.5 A PROJECT'S COST OF CAPITAL

Asset cost of capital or unlevered cost of capital

The expected return required by the firm's investors to hold the firm's underlying assets

The weighted average of the firm's assets equity and debt costs of capital

12.6 PROJECT RISK CHARACTERISTICS AND FINANCING

Firm asset betas reflect the market risk of the average project in a firm. But individual projects may be more or less sensitive to market risk.

Operating leverage

The relative proportion of fixed versus variable costs.

Weighted average cost of capital (WACC)

$$r_{WACC} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1 - \tau_c)$$

- ➔ The unlevered cost of capital is also referred to as the **pretax WACC**

CHAPTER 14: CAPITAL STRUCTURE IN A PERFECT MARKET

14.1 EQUITY VERSUS DEBT FINANCING

Capital structure

The relative proportions of debt, equity, and other securities that a firm has outstanding.

Unlevered equity

Equity in a firm with no debt

Levered equity

Equity in a firm that also has debt outstanding.

Promised payments to debt holders must be made before any payments to equity holders are distributed.

Leverage increases the risk of the equity of a firm. Investors in levered equity require a higher expected return to compensate for its increased risk.

➔ Leverage increases the risk of equity even when there is no risk that the firm will default

14.2 MODIGLIANI-MILLER: LEVERAGE, ARBITRAGE, AND FIRM VALUE

Perfect capital markets

1. Investors and firms can trade the same set of securities at competitive market prices equal to the present value of their future cash flows
2. There are no taxes, transaction costs, or issuance costs associated with security trading
3. A firm's financing decisions do not change the cash flows generated by its investments, nor do they reveal new information about them

MM Proposition I

In a perfect capital market, the total value of a firm's securities is equal to the market value of the total cash flows generated by its assets and is not affected by its choice of capital structure

In the absence of taxes or other transaction costs, the total cash flow paid out to all of the firm's security holders is equal to the total cash flow generated by the firm's assets. Therefore, by the Law of One Price, the firm's securities and its assets must have the same total market value.

➔ A firm's value isn't affected by its choice of capital structure

Homemade leverage

When investors use leverage in their own portfolios to adjust the leverage choice made by the firm.

As long as investors can borrow or lend at the same interest rate as the firm, homemade leverage is a perfect substitute for the use of leverage by the firm.

Market value balance sheet

It captures the idea that value is created by a firm's choice of assets and investments.

All assets and liabilities of the firm are included – even intangible assets that are missing from a standard accounting balance sheet.

All values are current market values rather than historical costs.

Market value of equity = Market value of assets – Market value of debt $\wedge$ other liabilities

Leveraged recapitalization

When a firm repurchases a significant percentage of its outstanding shares by borrowing

14.3 MM II: LEVERAGE, RISK AND THE COST OF CAPITAL

The MM Proposition I states that:

$$E + D = U = A$$

The total market value of the firm's securities is equal to the market value of its assets, whether the firm is unlevered or levered.

$$\frac{E}{E+D} R_E + \frac{D}{E+D} R_D = R_U$$

$$R_E = R_U + \frac{D}{E} (R_U - R_D)$$

The levered equity return equals the unlevered return, plus an extra “kick” due to leverage.

MM Proposition II

The cost of capital of levered equity increases with the firm's market value debt-equity ratio

Cost of Capital of Levered Equity

$$r_E = r_U + \frac{D}{E} (r_U - r_D)$$

Unlevered Cost of Capital (Pretax WACC)

$$r_U = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D = (\text{Fraction of Firm Value Financed by Equity})(\text{Equity Cost of Capital}) + (\text{Fraction of Firm Value Financed by Debt})(\text{Debt Cost of Capital})$$

- ➔ With perfect capital markets, a firm's WACC is independent of its capital structure and is equal to its equity cost of capital if it is unlevered, which matches the cost of capital of its assets
- ➔ The NPV of projects is unaffected by how a firm chooses to finance new investments

Debt-to-value ratio

The fraction of the firm's total value that corresponds to debt

$$\frac{D}{D+E}$$

A firm's unlevered or asset beta is the weighted average of its equity and debt beta

$$\beta_U = \frac{E}{E+D} \beta_E + \frac{D}{E+D} \beta_D$$

14.4 CAPITAL STRUCTURE FALLACIES

Leverage can increase a firm's expected EPS. But it will also increase risk.

The increase in the expected EPS is necessary to compensate shareholders for the additional risk they're taking on.

We can't reliably compare some measures across firms with different capital structures.

Dilution

If the firm issues new shares, the cash flows generated by the firm must be divided among a larger number of shares, thereby reducing the value of each individual share.
This ignores the fact that the cash raised by issuing new shares will increase the firm's assets.

CHAPTER 15 DEBT AND TAXES

15.1 THE INTEREST TAX DEDUCTION

A firm can be better off with leverage even though its earnings are lower.
Leverage allows the firm to pay out more in total to its investors, including interest payments to debt holders. So, it will be able to raise more total capital initially.

Interest tax shield

The additional amount that a firm would have paid in taxes if it didn't have leverage.
The gain to investors from the tax deductibility of interest payments.

$$\text{Interest Tax Shield} = \text{Corporate Tax Rate} * \text{Interest Payments}$$

15.2 VALUING THE INTEREST TAX SHIELD

Each year a firm makes interest payments, the cash flows it pays to investors will be higher than they would be without leverage by the amount of the interest tax shield

$$\text{Cash flows to investors with leverage} = \text{Cash flows to investors without leverage} + \text{interest tax shield}$$

MM Proposition I in the presence of taxes

The total value of the levered firm exceeds the value of the firm without leverage due to the present value of the tax savings from debt

$$V^L = V^U + PV(\text{Interest Tax Shield})$$

The interest tax shield with permanent debt

The firm maintains a fixed dollar amount of outstanding debt, rather than an amount that changes with the size of the firm

$$PV(\text{Interest Tax Shield}) = \frac{\tau_c * \text{Interest}}{r_f} = \frac{\tau_c * r_f * D}{r_f} = \tau_c * D$$

$$\text{Market value of debt} = D = PV(\text{Future Interest Payments})$$

Value of the Interest Tax Shield of Permanent Debt

$$PV(\text{Interest Tax Shield}) = PV(\tau_c * \text{Future Interest Payments}) = \tau_c * PV(\text{Future Interest Payments}) = \tau_c * D$$

The Weighted Average Cost of Capital with Taxes

When a firm uses debt financing, the cost of the interest it must pay is offset to some extent by the tax savings from the interest tax shield.

The tax deductibility of interest lowers the effective cost of debt financing for the firm

➔ With tax deductible interest, the effective after-tax borrowing rate is $r^*(1-\tau_c)$

WACC

$$r_{wacc} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1-\tau_c)$$

The higher the firm's leverage, the more the firm exploits the tax advantage of debt, and the lower its WACC is.

15.3 RECAPITALIZING TO CAPTURE THE TAX SHIELD

Recapitalization (recap)

When a firm makes a significant change to its capital structure

- ➔ When securities are fairly priced, the original shareholders of a firm capture the full benefit of the interest tax shield from an increase in leverage
- ➔ In presence of corporate taxes, we must include the interest tax shield as one of the firm's assets in the market value balance sheet

CHAPTER 16: FINANCIAL DISTRESS, MANAGERIAL INCENTIVES, AND INFORMATION

16.1 DEFAULT AND BANKRUPTCY IN A PERFECT MARKET

Default

A firm that fails to make the required interest or principal payments on the debt.

After the firm defaults, debt holders are given certain rights to the assets of the firm. In the extreme case, the debt holders take legal ownership of the firm's assets through a process called bankruptcy.

Economic distress

A significant decline in the value of a firm's assets, whether or not it experiences financial distress due to leverage.

16.2 THE COSTS OF BANKRUPTCY AND FINANCIAL DISTRESS

The bankruptcy code

The US bankruptcy code was created to organize the process so that creditors are treated fairly and the value of the assets is not needlessly destroyed.

Chapter 7 liquidation

A trustee is appointed to oversee the liquidation of the firm's assets through an auction.

Chapter 11 reorganization

In the more common form of bankruptcy for large corporations, all pending collection attempts are automatically suspended, and the firm's existing management is given the opportunity to propose a reorganization plan.

While developing the plan, management continues to operate the business. The reorganization plan specifies the treatment of each creditor of the firm. In addition to cash payment, creditors may receive new debt or equity securities of the firm.

The value of the cash and securities is generally less than the amount each creditor is owed, but more than the creditors would receive if the firm were shut down immediately and liquidated. The creditors must vote to accept the plan, and it must be approved by the bankruptcy court. If an acceptable plan is not put forth, the court may ultimately force a chapter 7 liquidation of the firm.

Workout

When a financially distressed firm is successful at reorganizing directly with creditors. Given the substantial legal and other direct costs of bankruptcy, firms in financial distress can avoid filing for bankruptcy by first negotiating directly with creditors.

Prepackaged bankruptcy

A firm will first develop a reorganization plan with the agreement of its main creditors and then file chapter 11 to implement the plan.

With a prepack, the firm emerges from bankruptcy quickly and with minimal direct costs.

Indirect costs of bankruptcy

Aside from the direct legal and administrative costs of bankruptcy, many other indirect costs are associated with financial distress (whether or not the firm has formally filed for bankruptcy)

- Loss of customers
- Loss of suppliers
- Loss of employees
- Loss of receivables
- Fire sale of assets
- Inefficient liquidation
- Costs to creditors

Debtor in possession (DIP) financing

New debt issued by a bankruptcy firm. This kind of debt is senior to all existing creditors, so it allows a firm that has filed for bankruptcy renewed access to financing to keep operating.

16.3 FINANCIAL DISTRESS COSTS AND FIRM VALUE

- ➔ When securities are fairly priced, the original shareholders of a firm pay the present value of the costs associated with bankruptcy and financial distress.

16.4 OPTIMAL CAPITAL STRUCTURE: THE TRADE-OFF THEORY

Trade-off theory

It weighs the benefits of debt that result from shielding cash flows from taxes against the costs of financial distress associated with leverage.

- ➔ The total value of a levered firm equals the value of the firm without leverage plus the present value of the tax savings from debt, less the present value of financial distress costs.

$$V^L = V^U + PV(\text{Interest tax shield}) - PV(\text{Financial Distress Costs})$$

The present value of financial distress costs

Three key factors determine the present value of financial distress costs:

1. The probability of financial distress
2. The magnitude of the costs if the firm is in distress
3. The appropriate discount rate for the distress costs

16.5 EXPLOITING DEBT HOLDERS: THE AGENCY COSTS OF LEVERAGE

Agency costs

Costs that arise when there are conflicts of interest between stakeholders.

- ➔ When a firm faces financial distress, shareholders can gain from decisions that increase the risk of the firm sufficiently, even if they have a negative NPV

Asset substitution problem

Leverage gives shareholders an incentive to replace low-risk assets with riskier ones

- ➔ When a firm faces financial distress, it may choose not to finance new, positive NPV projects

Debt overhang or under-investment problem

When shareholders prefer not to invest in a positive NPV project

Equity holders will benefit from the new investment only if

$$\frac{NPV}{I} > \frac{\beta_D D}{\beta_E E}$$

The project's profitability index must exceed a cutoff equal to the relative riskiness of the firm's debt times its debt-equity ratio

Leverage ratchet effect

Once existing debt is in place:

1. Shareholders may have an incentive to increase leverage even if it decreases the value of the firm
2. Shareholders will not have an incentive to decrease leverage by buying back debt, even if it will increase the value of the firm

The leverage ratchet effect is an important additional agency cost of leverage which affects the firm's future financing decisions, rather than its investment decisions.

While it'll induce firms to borrow less initially in order to avoid these costs, over time it may lead to excessive leverage as shareholders prefer to increase, but not decrease, the firm's debt.

Debt covenants

Restrictions on the actions that the firm can take. Covenants may limit the firm's ability to pay large dividends or restrict the types of investments that the firm can make.

16.6 MOTIVATING MANAGERS: THE AGENCY BENEFITS OF LEVERAGE

Management entrenchment

The separation of ownership and control creates the possibility of managers being free to run the firm in their own best interests.

Empire building

Managers prefer to run large firms rather than small ones, so they will take on investments that increase the size, rather than the profitability, of the firm.

Free cash flow hypothesis

The view that wasteful spending is more likely to occur when firms have high levels of cash flow in excess of what is needed to make all positive NPV investments and payments to debt holders.

16.7 AGENCY COSTS AND THE TRADE OFF THEORY

$$V^L = V^U + PV(\text{Interest Tax Shield}) - PV(\text{Financial Distress Costs}) - PV(\text{Agency Costs of Debt}) + PV(\text{Agency Benefits of Debt})$$

The optimal level of debt, D^* , balances the costs and benefits of leverage

The tradeoff theory explains how firms should choose their capital structures to maximize value to current shareholders.

Managers seek to minimize leverage to prevent the job loss that would accompany financial distress.

16.8 ASYMMETRIC INFORMATION AND CAPITAL STRUCTURE

The credibility principle

Claims in one's self-interest are credible only if they are supported by actions that would be too costly to take if the claims were untrue.

- ➔ Managers want to give off signals that the firm is doing good, like press releases that are able to be compared after the fact, or committing to large future payments

The signaling theory of debt

The use of leverage as a way to signal good information to investors.

Adverse selection

The selection is worse than average because everyone thinks the other is screwing them over

The lemons principle

When a seller has private information about the value of a good, buyers will discount the price they're willing to pay due to adverse selection

There's a preference not to issue equity that's underpriced

- ➔ Managers who know securities have a high value won't sell, and those who know they have a low value will sell

The lemons problem creates a cost for firms that need to raise capital from investors to fund new investments. If they try to issue equity, investors will discount the price they're willing to pay to reflect the possibility that managers are privy to bad news.

Implications for equity issuance

1. **The stock price declines on the announcement of an equity issue.** When a firm issues equity, it signals to investors that its equity may be overpriced. As a result, investors aren't willing to pay the pre-announcement price for the equity and so the stock price declines.
2. **The stock price tends to rise prior to the announcement of an equity issue.**
3. **Firms tend to issue equity when information asymmetries are minimized, such as immediately after earnings announcements**

- ➔ Managers who perceive the firm's equity is underpriced will have a preference to fund investment using retained earnings, or debt, rather than equity.

Pecking order hypothesis

Managers will prefer to use retained earnings first and will issue new equity only as a last resort.

CHAPTER 17: PAYOUT POLICY

17.1 DISTRIBUTIONS TO SHAREHOLDERS

Payout policy

The way a firm chooses between retaining and paying out free cash flow

Declaration date

The date on which the board authorizes the dividend

Record date

The specific date set by the board in which the firm will pay the dividend to all shareholders of record

Ex-dividend date

The date two business days prior to the record date

Payable/distribution date

The firm mails dividend checks to the registered shareholders

Special dividend

A one time dividend that's much larger than a regular one

Stock split or stock dividend

The company issues additional shares rather than cash to its shareholders.

Return of capital or liquidating dividend

When the dividends are attributed to other accounting sources, such as paid-in capital or the liquidation of assets

Share repurchases

An alternative way to pay cash to investors is through a share repurchase or buyback. The firm uses cash to buy shares of its own outstanding stock.

Open market repurchase

A firm announces its intention to buy its own shares in the open market, and then proceeds to do so over time like any other investor.

Tender offer

The firm offers to buy shares at a prespecified price during a short time period.

Dutch auction

The firm lists different prices at which it's prepared to buy shares, and shareholders in turn indicate how many shares they're willing to sell at each price. The firm then pays the lowest price at which it can buy back its desired number of shares

Targeted repurchase

The purchase price is negotiated directly with the seller.

Greenmail

When a major shareholder is threatening to take over the firm and remove its management, the firm may decide to eliminate the threat by buying out the shareholder

17.2 COMPARISON OF DIVIDENDS AND SHARE REPURCHASES

In the perfect capital markets setting of Modigliani and Miller, the method of payment (between dividends and repurchases) doesn't matter.

Alternative #1: pay dividends with excess cash

Cum-dividend

The stock is said to trade cum-dividend (with dividend) just before the ex-dividend trade because anyone who buys the stock will be entitled to the dividend

$$P_{cum} = \text{Current dividend} + PV(\text{Future dividends})$$

After the stock goes ex-dividend, new buyers won't receive the current dividend

$$P_{ex} = PV(\text{Future dividends})$$

The share price will drop after the ex-dividend date

The share price falls when a dividend is paid because the reduction in cash decreases the market value of the firm's assets. But the shareholders don't incur a loss because they received the dividend.

- ➔ In a perfect capital market, when a dividend is paid, the share price drops by the amount of the dividend when the stock begins to trade ex-dividend

Alternative #2: share repurchase (no dividend)

The market value of the firm falls when the company pays out cash, but the number of shares outstanding also falls. The two changes offset each other, so the share price remains the same.

Homemade dividend

If the firm repurchases shares and the investor wants cash, they can raise cash by selling shares.

- ➔ In perfect capital markets, investors are indifferent between the firm distributing funds via dividends or share repurchases. By reinvesting dividends or selling shares, they can replicate either payout method on their own.

Alternative #3: high dividend (equity issue)

The initial share value is unchanged and increasing the dividend has no benefit to shareholders

MM dividend irrelevance

In perfect capital markets, holding fixed the investment policy of a firm, the firm's choice of dividend policy is irrelevant and doesn't affect the initial share price.

17.3 THE TAX DISADVANTAGE OF DIVIDENDS

Shareholders typically must pay taxes on the dividends they receive. They must also pay capital gains taxes when they sell their shares.

If dividends are taxed at a higher rate than capital gains, which was true prior to 2003, shareholders will prefer share repurchases to dividends. And although recent tax code changes

equalized the tax rates on dividends and capital gains, because capital gains taxes are deferred until the asset is sold, there is still a tax advantage for share repurchases over dividends for long-term investors.

A higher tax rate on dividends also makes it undesirable for a firm to raise funds to pay a dividend. When dividends are taxed at a higher rate than capital gains, this transaction hurts shareholders because they'll receive less than their initial investment.

Optimal dividend policy with taxes

Pay no dividends at all.

Dividend puzzle

The fact that firms continue to issue dividends despite their tax disadvantage

17.4 DIVIDEND CAPTURE AND TAX CLIENTELES

Tax rates vary by income, jurisdiction, investment horizon, and whether the stock is held in a retirement account.

Because of these differences, firms may attract different groups of investors depending on their dividend policy.

The effective dividend tax rate

$$\tau_d^e = \left(\frac{\tau_d - \tau_g}{1 - \tau_g} \right)$$

τ_d : tax on dividends

τ_g : tax on capital gains

Tax differences across investors

The effective dividend tax rate for an investor depends on the tax rates the investor faces on dividends and capital gains. These rates differ across investors for a variety of reasons:

- **Income level**
- **Investment horizons:** capital gains on stocks held one year or less, and dividends on stocks held for less than 61 days, are taxed higher ordinary income tax rates. Long-term investors can defer the payment of capital gains taxes, lowering their effective capital gains tax rate even further. Investors who plan to bequeath stocks to their heirs may avoid the capital gains tax altogether.
- **Tax jurisdiction:** US investors are subject to state taxes that differ by state.
- **Type of investor or investment account:** stocks held by individual investors in a retirement account or through pension funds or nonprofit endowment funds aren't subject to taxes on dividends or capital gains. Corporations that hold stocks are able to exclude 70% of dividends they receive from corporate taxes but are unable to exclude capital gains

Clientele effects

Differences in tax preferences across investor groups create clientele effects, in which the dividend policy of a firm is optimized for the tax preference of its investor clientele.

Dividend-capture theory

Absent transaction costs, investors can trade shares at the time of the dividend so that non-taxed investors receive the dividend.

That is, non-taxed investors don't need to hold high-dividend paying stocks all the time; it's necessary only that they hold them when the dividend is actually paid.

17.5 PAYOUT VERSUS RETENTION OF CASH

How should the firm decide the amount it should pay to shareholders and the amount it should retain?

Retaining cash with perfect capital markets

If it retains, it can use those funds to invest in new projects.

If new $NPV > 0$ projects are available, this decision is clearly the correct one.

Once the firm has taken all $NPV > 0$ projects, any additional projects it takes on will be $NPV \leq 0$ and this will reduce shareholder value.

Instead, a firm can hold the cash in the bank or use it to purchase financial assets.

In perfect capital markets, buying and selling securities is an $NPV = 0$ transaction, so it shouldn't affect firm value. Shareholders can make any investment a firm makes on their own if the firm pays out the cash.

MM payout irrelevance

In perfect capital markets, if a firm invests excess cash flows in financial securities, the firm's choice of payout vs retention is irrelevant and doesn't affect the initial value of the firm.

Taxes and cash retention

Corporate taxes make it costly for a firm to retain excess cash.

Cash is equivalent to negative leverage, so the tax advantage of leverage implies a tax disadvantage to holding cash.

There's one situation where holding cash can reduce taxes. Corporations only pay US taxes on their international earnings when they repatriate those earnings, so they can lower their taxes by not repatriating earnings and instead holding the cash abroad.

Price an investor will pay for a firm that retains

$$P_{\text{retain}} = P_{\text{cum}} (1 - \tau_{\text{retain}})$$

Effective tax disadvantage of retaining cash (for investors)

$$\tau_{retain}^i = \left(1 - \frac{(1-\tau_c)(1-\tau_g)}{1-\tau_i} \right)$$

Issuance and distress costs

Firms retain cash balances even with a tax disadvantage to cover potential future cash shortfalls.

Example: if there's a likelihood that future earnings will be insufficient to fund future NPV>0, a firm may start accumulating cash to make up the difference.

This strategy allows a firm to avoid the transaction costs of raising new capital.

By holding sufficient cash, these firms can avoid financial distress and its associated costs

Agency costs of retaining cash

There's no benefit to shareholders when a firm holds cash above and beyond its future investment or liquidity needs.

When firms have excessive cash, managers may use the funds inefficiently by continuing money-losing per projects, paying excessive executive perks, or over-paying for acquisitions. Also, unions, the government, or other entities may take advantage.

Ultimately, firms should choose to retain cash to preserve financial slack for future growth opportunities and to avoid financial distress costs. These needs must be balanced against the tax disadvantage of holding cash and the agency cost of wasteful investment.

17.6 SIGNALING WITH PAYOUT POLICY

Dividend smoothing

The practice of maintaining relatively constant dividends.

Firms also increase dividends much more frequently than they cut them.

Firms keep dividends smooth even if earnings vary by adjusting the number of shares they repurchase or issue and the amount of cash that they retain.

Due to tax and transactions costs, firms also set dividends at a level they expect to be able to maintain based on the firm's earnings prospects

Dividend signaling hypothesis

The idea that dividend changes reflect managers' views about a firm's future earnings prospects.

When a firm increases its dividends, it sends a positive signal to investors that management expects to be able to afford the higher dividend for the foreseeable future. However, it may also signal a lack of investment opportunities.

When managers cut dividends, it may signal that they've given up hope that earnings will rebound in the near term and so need to reduce the dividend to save cash. A firm may also cut dividends to exploit new NPV>0 investment opportunities, which can lead to a positive stock price reaction.

So, dividend signaling is like debt signaling, but while cutting the dividend is costly for managers in terms of their reputation and the reaction of the investors, it's not as costly as failing to make debt payments. So, we'd expect that dividend signaling to be weaker than debt signaling, but empirical studies show otherwise.

Signaling and share repurchases

Share repurchases may also signal managers' information to the markets. However, there are several important differences.

- Managers are less committed to share repurchases than to dividend payments. Firms generally announce the maximum amount they plan to spend, but the actual amount may be far less. It also may take several years to complete the share repurchase
- Firms don't smooth their repurchase activity from year to year. So, announcing a repurchase today doesn't necessarily represent a long-term commitment to repurchase shares. So, share repurchases may be less of a signal than dividends about future earnings of a firm
- The cost of a share repurchase depends on the market price of the stock

Share repurchases are a credible signal that management believes its shares are underpriced. Since investors believe that managers have better information regarding the firm's prospects than they do, they react favorably to share repurchase announcements.

CHAPTER 18: CAPITAL BUDGETING AND VALUATION WITH LEVERAGE

18.2 THE WEIGHTED AVERAGE COST OF CAPITAL METHOD

$$r_{wacc} = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D (1 - \tau_c)$$

1. Determine the free cash flow of the investment
2. Compute the weighted average cost of capital using the above equation
3. Compute the value of the investment, including the tax benefit of leverage, by discounting the free cash flow of the investment using the WACC

➔ The WACC can be used throughout the firm as the companywide cost of capital for new investments that are of comparable risk to the rest of the firm and that will not alter the firm's debt-equity ratio

Debt capacity

The amount of debt at date t that's required to maintain the firm's target debt-to-value ratio

18.3 THE ADJUSTED PRESENT VALUE METHOD

An alternative valuation method in which we determine the levered value V^L of an investment by first calculating its unlevered value V^U , which is its value without any leverage, and then adding the value of the interest tax shield.

The APV Formula

$$V^L = APV = V^U + PV(\text{Interest Tax Shield})$$

The project's cost of capital if it were financed without leverage

$$r_U = \frac{E}{E+D} r_E + \frac{D}{E+D} r_D = \text{Pretax WACC}$$

Target leverage ratio

The firm adjusts its debt proportionally to the project's value, or its cash flows, so that a constant debt-equity ratio is a special case

The interest tax shield is equal to the interest paid multiplied by the corporate tax rate

- ➔ When the firm maintains a target leverage ratio, its future interest tax shields have similar risk to the project's cash flows, so they should be discounted at the project's unlevered cost of capital
1. Determine the investment's value without leverage, V^U , by discounting its free cash flows at the unlevered cost of capital, r_U . With a constant debt-equity ratio, r_U , may be estimated.
 2. Determine the present value of the interest tax shield
 - a. Determine the expected interest tax shield: Given expected debt D_t on date t , the interest tax shield on date $t+1$ is $\tau_c r_D D_t$
 - b. Discount the interest tax shield. If a constant debt-equity ratio is maintained, using r_U is appropriate
 3. Add the unlevered value, V^U , to the present value of the interest tax shield to determine the value of the investment with leverage, V^L .

CHAPTER 20: FINANCIAL OPTIONS

20.1 OPTION BASICS

Financial option

Gives its owner the right, but not the obligation, to purchase or sell an asset at a fixed price at some future date.

Two types exist: call options and put option

Call option

Gives the owner the right to buy the asset

Put option

Gives the owner the right to sell the asset

Exercising the option

When the holder of an option enforces the agreement and buys or sells a share of stock at the agreed-upon price

Strike price / Exercise price

The price at which the holder buys or sells the share of stock when the option is exercised

American options

The most common kind

They allow their holders to exercise the option on any date up to and including a final date called the **expiration date**

European options

They allow their holders to exercise the option only on the expiration date

- ➔ Both types of options are traded worldwide
- ➔ The short side has an *obligation* to fulfill the contract

Why do people write (sell) options?

- When you sell an option, you get paid for it.
- You get an upfront payment that compensates the seller for the risk of loss in the event that the option holder chooses to exercise the option
- Options always have positive prices

Option premium: the market price of the option

Open interest

The total number of outstanding contracts of that option

At-the-money

When the exercise price of an option is equal to the current price of the stock

In-the-money

The payoff from exercising an option immediately is positive

Out-of-the-money

The payoff from exercising the option immediately is negative

Deep in-the-money or Deep out-of-the-money

Options where the strike price and the stock price are very far apart

Hedging

Using options to offset the losses on a portfolio

Speculate

Place a bet on the direction in which they believe the market is likely to move

➔ Options allow investors to hedge or speculate

20.2 OPTION PAYOFFS AT EXPIRATION

Call value at expiration

$$C = \max(S - K, 0)$$

Put price at expiration

$$P = \max(K - S, 0)$$

Combinations of options

- **Straddle:** you're long on both a put option and a call option with the same strike price. You'll receive cash as long as the options don't expire at-the-money. The farther away from the money the options are, the more money you'll make.
- **Strangle:** a call and a put with the same expiration date but one has a higher exercise price. You don't receive money if the stock price is between the two strike prices.
- **Butterfly spread:** long on two call options with different strike price and the same expiration date, plus short on two call options with the same strike price.

Protective put

You buy a put on the same stock you hold in order to protect yourself against the possibility of a price decline.

Portfolio insurance

Insure against a loss on an entire portfolio of stocks by using put options on the portfolio of stocks

You can achieve the same effect by purchasing a bond and a call option.

20.3 PUT-CALL PARITY

Consider the two different ways to construct portfolio insurance:

1. Purchase the stock and a put
2. Purchase a bond and a call

Because both positions provide exactly the same payoff, the Law of One Price requires that they must have the same price.

Put call parity

$$S + P = PV(K) + C$$

The price of a European call equals the price of the stock plus an otherwise identical put minus the price of a bond that matures on the exercise date of the option

➔ You can think of a call as a combination of a levered position in the stock, $S - PV(K)$, plus insurance against a drop in the stock price, the put P

If the stock pays dividends

$$S + P = PV(K) + PV(D) + C$$

20.4 FACTORS AFFECTING OPTION PRICES

Strike price and stock price

The value of an otherwise identical call option is higher if the strike price is lower. Puts with a lower strike price are less valuable

For a given strike price, the value of a call option is higher if the current price of the stock is higher, as there's a greater likelihood the option will end up in-the-money.

Put options increase in value as the stock price falls

Arbitrage bounds on option prices

An option's price can't be negative.

An American option can't be worth less than a European one.

A put option can't be worth more than its strike price

A call option can't be worth more than the stock itself

An American option can't be worth less than its intrinsic value

Intrinsic value

The value an option would have if it expired immediately.

The amount by which the option is currently in the money, or zero if it's out of the money

Time value

The difference between the current option price and its intrinsic value

Option prices and the Exercise date

For American options, the longer the time to the exercise date, the more valuable the option.

An American option with a later exercise date can't be worth less than an otherwise identical American option with an earlier exercise date

➔ The same can't be said with European options

Option prices and volatility

The value of an option generally increases with the volatility of the stock

20.5 EXERCISING OPTIONS EARLY

Non-dividend paying stocks

The price of any call option on a non-dividend paying stock always exceeds its intrinsic value

➔ This implies that **it's never optimal to exercise a call option on a non-dividend stock early**

➔ You're always better off just selling the option

An American call on a non-dividend paying stock has the same price as its European counterpart

Dividend-paying stocks

Calls: the best time to exercise early is just before the dividend's announcement

Puts: just after the announcement. So, it's less likely that a put will be exercised early

20.6 OPTIONS AND CORPORATE FINANCE

Equity as a call option

Think of share of stock as a call option on the assets of the firm with a strike price equal to the value of debt outstanding.

If the firm's value doesn't exceed the value of debt outstanding at the end of the period, the firm must declare bankruptcy and the equity holders receive nothing. If the value exceeds the value of debt outstanding, the equity holders get whatever is left once the debt has been repaid.

Debt as an option portfolio

You can think of debt holders as owning the firm and having sold a call option with a strike price equal to the required debt payment.

If the value of the firm exceeds the required debt payment, the call will be exercised, and the debt holders will receive the strike price and give up the firm. If the value of the firm doesn't exceed the required debt payment, the call will be worthless, the firm will declare bankruptcy, and the debt holders will be entitled to the firm's assets.

Credit Default Swaps

Risk-free debt = Risky debt + Put option on firm assets

We can eliminate a bond's credit risk by buying the very same put option to protect or ensure it. We refer to this put option, which can insure a firm's credit risk, as a credit default swap (or CDs)

The buyer pays a premium to the seller (often in the form of periodic payments) and receives a payment from the seller to make up for the loss if the underlying bond defaults

Agency conflicts

Recall that the price of an option generally increases with the volatility level of the underlying security.

Because equity is like a call option on the firm's assets, equity holders will benefit from investments that increase the risk of the firm. On the other hand, debt holders are short a put option on the firm's assets. Thus, they'll be hurt by an increase in the firm's risk.

This conflict of interest regarding risk-taking is the asset substitution problem, and we can quantify it in terms of the sensitivity of the option values to the firm's volatility

When a firm makes new investments that increase the value of the firm's assets, the value of a put option on the firm will decline. Because debt holders are short a put option, the value of the firm's debt will increase. Thus, some fraction of each dollar increase in the value of the firm's assets will go to debt holders, rather than equity holders, reducing equity holder's incentive to invest. This problem is the debt overhang problem.

CHAPTER 21: OPTION VALUATION

21.1 THE BINOMIAL OPTION PRICING MODEL

The model prices options by making the simplifying assumption that at the end of the next period, the stock price has only two possible values.

- ➔ Option payoffs can be replicated exactly by constructing a portfolio out of a risk-free bond and the underlying stock

Replicating portfolio

A portfolio of other securities that has exactly the same value in one period as the option. Because they have the same payoffs, the Law of One Price implies that the current value of the call and the replicating portfolio must be equal.

Binomial tree

A timeline with two branches at every date representing the possible events that could happen at those times

- ➔ By using the Law of One Price, we're able to solve for the price of the option without knowing the probabilities of the states in the binomial tree

Replicating portfolio in the binomial model

$$\Delta = \frac{C_u - C_d}{S_u - S_d}$$
$$B = \frac{C_u - S_u \Delta}{1 + r_f}$$

Option price in the Binomial Model

$$C = S\Delta + B$$

Making the model realistic

By decreasing the length of each period, and increasing the number of period in the stock price tree, we can construct a realistic model for the stock price

21.2 THE BLACK-SCHOLES OPTION PRICING MODEL

It can be derived from the Binomial Option Pricing Model by making the length of each period, and the movement of the stock price per period, shrink to zero and letting the number of periods grow infinitely large.

The Black-Scholes formula

S: current price of the stock

T: number of years left to expiration

K: exercise price

Sigma: the annual volatility (standard deviation) of the stock's return

The value, at time t , of a call option on a stock that doesn't pay dividends prior to the option's expiration date is given by

Black-Scholes price of a call option on a non-dividend paying stock

$$C = S * N(d_1) - PV(K) * N(d_2)$$

Where $PV(K)$ is the present value of a risk-free zero-coupon bond that pays K on the expiration date of the option

And $N(d)$ is the **cumulative normal distribution**, the probability, that a normally distributed variable is less than d , and

$$d_1 = \frac{\ln\left[\frac{S}{PV(K)}\right] + \frac{\sigma\sqrt{T}}{2}}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

We need only five inputs to price the call:

1. Stock price
2. Strike price
3. Exercise date
4. The risk-free interest rate
5. Volatility of the stock (only one not directly observable)

The Black-Scholes formula is derived assuming that the call is a European option

- An American call option on a non-dividend paying stock always has the same price as its European counterpart

Black-Scholes price of a European put option on a Non-Dividend paying stock

$$P = C - S + PV(K)$$

$$P = PV(K)[1 - N(d_2)] - S[1 - N(d_1)]$$

Dividend paying stocks

The holder of a European call option doesn't receive the benefit of any dividends that'll be paid prior to the expiration date of the option. The stock price tends to drop by the amount of the dividend when the stock goes ex-dividend. Because the final stock price will be lower, dividends decrease the value of a call option.

PV(DIV): present value of any dividends paid prior to the expiration date of the option.

A security that's identical to the stock, but that didn't pay any dividends would have a current market price of

$$S^x = S - PV(i)$$

The value of S^x is the current price of the stock excluding any dividends prior to expiration.

- ➔ Because a European call option is the right to buy the stock without these dividends, we can evaluate it using the Black-Scholes formula with S^x in place of S

A useful special case is when the stock will pay a dividend that's proportional to its stock price at the time the dividend is paid. If q is the stock's dividend yield until the expiration date, then

$$S^x = \frac{S}{1+q}$$

Approaches to estimate the value of the volatility of the stock

1. Use historical data on daily stock returns to estimate volatility
2. Use current market prices of traded options to "back out" the volatility that's consistent with these prices based on the Black-Scholes formula.

Implied volatility

An estimate of a stock's volatility that's implied by an option's price.

The implied volatility from one option can be used to estimate the value of other options on the stock with the same expiration date.

The replicating portfolio

The price of a call option is given by the price of the replicating portfolio

$$C = S\Delta + B$$

Black-Scholes replicating portfolio of a call option

$$\Delta = N(d_1)$$

$$B = -PV(K) * N(d_2)$$

The option delta (Δ)

Natural interpretation: it's the change in the price of the option given a \$1 change in the price of the stock.

The change in the call price is always less than the change in the stock price.

The replicating portfolio of a call option always consists of a long position in the stock and a short position in the bond

- ➔ The replicating portfolio is a leveraged position in the stock

Because a leveraged position is riskier than the stock itself, this implies that call options on a positive beta stock are more risky than the underlying stock and therefore have higher returns and higher betas.

Black-Scholes replicating portfolio of a put option

$$\Delta = -[1 - N(d_1)]$$

$$B = PV(K)[1 - N(d_2)]$$

The replicating portfolio of a put option always consists of a long position in the bond and a short position in the stock, implying that put options on a positive beta stock will have a negative beta.

21.5 CORPORATE APPLICATIONS OF OPTION PRICING

Beta of risky debt

Before, we assumed that the beta of debt was zero, so

$$\beta_E = \beta_U + \frac{D}{E}(\beta_U - \beta_D) \approx \left(1 + \frac{D}{E}\right) \beta_U$$

However, for companies with high equity-to-debt ratios, the approximation that the beta of debt is zero is unrealistic.

Because equity is a call option on the assets of the firm, we can write the value of equity as:

$$E = A\Delta + B$$

A: the value of the firm's assets.

A=E+D is used in place of S to represent the underlying assets on which the option is written

$$\beta_E = \frac{A\Delta}{A\Delta + B} \beta_U = \frac{(E+D)\Delta}{E} \beta_U = (1-\Delta)\left(1 + \frac{D}{E}\right) \beta_U$$

We can derive the beta of debt in a similar fashion

$$D = A - E$$

$$\beta_D = \frac{A}{D} \beta_U - \frac{E}{D} \beta_E$$

$$\beta_D = (1-\Delta) \frac{A}{D} \beta_U = (1-\Delta)\left(1 + \frac{E}{D}\right) \beta_U$$

$$\beta_U = \frac{\beta_E}{\Delta\left(1 + \frac{D}{E}\right)}$$

Agency costs of debt

Leverage creates an assets substitution problem, because the value of the equity call option increases with the firm's volatility. Thus, equity holders may have an incentive to take excessive risk.

Because $\Delta < 1$ for a call option, equity holders gain less than \$1 for each \$1 increase in the value of the firm's assets, reducing their incentive to invest and leading to a debt overhand or underinvestment problem

COMMON MISTAKE: VALUING EMPLOYEE STOCK OPTIONS

It's become common practice to compensate executives by granting them **executive stock options (ESOs)**, call options on their company's stock.

ESOs are typically American-style options with exercise dates up to 10 years in the future. There is usually a vesting period during which the employee doesn't actually own the option. Instead, they own a right to the option at the end of the vesting period. If the employee leaves the firm during this period, the individual forfeits this right and so doesn't get the option.

Once the vesting period has passed, the employee owns the option, but it isn't tradeable. The only way the employee can liquidate the option is by exercising it. Because of these restrictions, ESOs aren't worth the same amount to the employee and the firm.

Difficulties when trying to apply Black-Scholes to employee stock options:

1. The formula requires an estimate of volatility over the life of the option.
2. The formula doesn't account for important differences between ESOs and ordinary options:
 - a. ESOs are dilutive: when exercised, they increase the number of shares
 - b. ESOs may be forfeited
 - c. ESOs may be exercised early

Employees are risk averse but aren't permitted to hedge the risk of the option by trading the replicating portfolio. As a consequence, the employee's beliefs matter in computing the ESO's value: a more risk-averse employee will attach a lower value

The only way to eliminate the risk from the option is to exercise it and sell the stock

➔ Most employees choose to exercise early

Thus, the Black-Scholes formula overestimates the cost of the option to the firm and its benefit to the employee.

CHAPTER 22: REAL OPTIONS

22.1 REAL VERSUS FINANCIAL OPTIONS

The financial options we've studied in the previous chapters give their holders the right to buy, or sell, a traded asset such as a stock.

Real option

The right to make a particular business decision, such as a capital investment, after new information may be learned.

Real options, and the underlying assets, aren't often traded in competitive markets.

The presence of real options adds value to an investment opportunity

22.2 DECISION TREE ANALYSIS

Decision tree

A graphical representation that shows current and future decisions and their corresponding risks and outcomes over time

➔ Difference with a binomial tree: we also include branches to represent different choices available to the decision maker

Types of nodes

- **Decision nodes:** invest versus do nothing
- **Information nodes:** in which uncertainty is resolved that is out of control of the decision maker
- ➔ **A decision node that occurs after an information node in a tree is a real option**

Solving decision trees

Real options can be analyzed by creating a decision tree that identifies:

- Decision nodes showing the choices available at each stage
- Information nodes showing the payoff relevant information to be learned
- Investments made and payoffs earned over time

Once the decision tree is created, we can value the investment opportunity by working backward from the end of the tree. At each:

1. **Decision node:** determine the optimal choice by comparing the present value of the remaining payoffs along each branch
2. **Information node:** compute the expected present value of the payoffs from the subsequent branches

22.3 THE OPTION TO DELAY: INVESTMENT AS A CALL OPTION

Cons of delaying an investment:

- Interim profits from the project are lost
- Costs might rise
- Competitors may enter

Pros of delaying an investment:

- Gain additional information regarding the value of the investment

Trade off: costs of the delay with the benefit from gaining information before making a decision

Factors affecting the timing of investment

Without the option of when to invest, it's optimal to invest as long as $NPV > 0$

- ➔ But when you have the option of deciding when to invest, it's usually optimal to invest only when the NPV is substantially greater than zero

Think of the timing decision as a choice between two mutually exclusive projects: invest today or wait. Faced with mutually exclusive choices, we should choose the project with the higher NPV. We should invest today only if the NPV of investing today exceeds the value of the option of waiting.

- ➔ Given the option to wait, an investment that currently has a negative NPV can have a positive value

Other factors:

- **Volatility:** the option to wait is most valuable when there's a great deal of uncertainty regarding what the value of the investment will be in the future. If there's little uncertainty, the benefit of waiting is diminished
 - **Dividends:** in the real options context, the dividends correspond to any value from the investment that we give up by waiting. It's always better to wait unless there's a cost to doing so. The greater the cost, the less attractive the option to delay becomes
- ➔ All else equal, firms for which a higher fraction of their value depends on future growth will tend to have higher betas

22.4 GROWTH AND ABANDONMENT OPTIONS

Growth option

When a firm has a real option to invest in the future

Abandonment option

When the firm has the option to reduce the scale of its investment in the future

CHAPTER 28: MERGERS AND ACQUISITIONS

28.1 BACKGROUND AND HISTORICAL TRENDS

Merger waves

Peaks of heavy activity followed by quiet troughs of few transactions

Horizontal merger

The target and acquirer are in the same industry

Vertical merger

The target's industry buys or sells to the acquirer's industry

Conglomerate merger

The target and acquirer operate in unrelated industries

Stock swap

When the target shareholders receive stock as payment for target shares.

They're swapping their old stock for new stock in either the acquirer or the newly created merged firm

Term sheet

Summary of the structure of a merger transaction

28.2 MARKET REACTION TO A TAKEOVER

A bidder is unlikely to acquire a target company for less than its current market value. Instead, most acquirers pay a substantial acquisition premium

Acquisition premium

The percentage difference between the acquisition price and the premerger price of the target firm

28.3 REASONS TO ACQUIRE

Why can an acquirer pay a premium and still get $NPV > 0$?

An acquirer might be able to add economic value, as a result of the acquisition, that an individual investor might not add.

Large synergies are the most common justification that bidders give for the premium they pay for a target.

They usually fall into two categories:

- Cost-reduction synergies
- Revenue enhancements

Cost-reduction synergies

More common and easier to achieve because they generally translate into layoffs of overlapping employees and elimination of redundant resources.

Economies of scale

Savings from producing goods in high volume

Economies of scope

Savings that come from combining the marketing and distribution of different types of related products.

Vertical integration

A merger of two companies in the same industry that make products required at different stages of the production cycle.

The principal benefit is coordination.

Expertise

An efficient solution to needing expertise may be to purchase the talent as an already functioning unit by acquiring an existing firm.

Monopoly gains

It's often argued that merging with or acquiring a major rival enables a firm to substantially reduce competition within the industry and thereby increase profits. Most countries have antitrust laws that limit these activities.

We'd expect that in the absence of strong antitrust laws, many companies would merge. However, while all companies in the industry benefit when competition is reduced, only the merging company pays the associated costs.

Efficiency gains

Often achieved through an elimination of duplication

Tax savings from operating losses

The IRS will disallow a tax break if it can show that the principal reason for a takeover is tax avoidance

Diversification

- **Risk reduction:** this argument ignores the fact that investors can achieve the benefits of diversification themselves by purchasing shares in two separate firms. It's cheaper for investors to diversify their own portfolios than to have the corporation do it through acquisition
- **Debt capacity and borrowing costs:** all else equal, larger, more diversified firms have a lower probability of bankruptcy given the same degree of leverage. They can also increase leverage further and enjoy greater tax savings without incurring significant costs of financial distress.
- **Liquidity:** shareholders of private companies are often under-diversified. When an acquirer buys a private target, it provides the target's owners a way to reduce their risk by cashing out.

Earning growth

It's possible to combine two companies with the result that the earnings per share of the merged company exceed the premerger earnings per share of either company, even when the merger itself creates no economic value.

Managerial motives to merge

Studies have found that the stock price of large bidders drops on average when a bid is announced, especially when the target is publicly traded.

Two possible explanations:

- **Conflicts of interest with their shareholders:** managers may prefer to run a larger company due to the additional pay and prestige it brings. Because most CEOs hold only a small fraction of their firm's stock, they may not bear enough of the cost of an otherwise bad merger that increases their personal benefits.
- **Overconfidence:** Roll's "hubris hypothesis": overconfident CEOs pursue mergers that have low chance of creating value because they truly believe that their ability to manage is great enough to succeed.
- ➔ **Difference between the two:** overconfident managers believe they're doing the right thing for their shareholders, but irrationally overestimate their own abilities. Under the incentive conflict explanation, managers know they're destroying shareholder value, but personally gain from doing so.

28.4 VALUATION AND THE TAKEOVER PROCESS

Valuation

Two approaches:

- Comparing the target to other comparable companies
- Making a projection of the expected cash flows that will result from the deal, and valuing those cash flows

A key issue for takeovers is quantifying and discounting the value added as a result of the merger.

$$\text{Amount paid} = \text{Target's PreBid Market Capitalization} + \text{Acquisition premium}$$

From the bidder's standpoint, the benefit from the acquisition includes both the standalone value of the target and any synergies created

$$\text{Value acquired} = \text{Target standalone value} + PV(\text{Synergies})$$

Tender offer

A public announcement of its intention to purchase a large block of shares for a specified price

Exchange ratio

The number of bidder shares received in exchange for each target share

Condition for a stock-swap merger to have NPV>0

$$\frac{A+T+S}{N_A+X} > \frac{A}{N_A} = P_A$$

$$XP_A < T+S$$

A= premerger or standalone value of the acquirer

T= premerger or standalone value of the target

S= value of synergies created by the merger

NA= number of shares outstanding before the merger (acquirer)

PA= share price of each NA before the merger

X= number of new shares issued

$$\text{Exchange ratio} = \frac{X}{N_T} < \frac{P_T}{P_A} \left(1 + \frac{S}{T}\right)$$

PT=T/NT= premerger value of the target

Once a tender offer is announced, there's no guarantee that the takeover will take place at this price or even at all.

This uncertainty creates an opportunity for investors to speculate on the outcome of the deal.

Risk arbitrageurs

Traders who believe they can predict the outcome of a deal

They take positions based on their beliefs.

Even if this is "arbitrage", it's actually quite risky.

Merger-arbitrage spread

The difference between the target's stock price and the implied offer price

Once the terms of the trade have been decided, the tax and accounting implications of a merger can be determined.

For a takeover to proceed, both the target and the acquiring board of directors must approve the deal and put the question to a vote of the shareholders of the target. In some cases, of the acquirer as well

Friendly takeover

The target board of directors supports the merger, negotiates with potential acquirers and agrees on a price that's ultimately put to a shareholder vote

Hostile takeover

The board of directors fights the takeover attempt

To succeed, the acquirer must garner enough shares to take control of the target and replace the board of directors.

When a takeover is hostile, the acquirer is often called a **raider**

Why oppose a takeover?

- The board thinks the price is too low
- The board thinks the stock offered is over-valued
- Managers might oppose a takeover for personal interests

28.5 TAKEOVER DEFENSES

Proxy fight

The acquirer attempts to convince target shareholders to unseat the target board by using their proxy votes to support the acquirer's candidates for election to the target board.

Poison pill

A rights offering that gives existing target shareholders the right to buy shares in the target at a deeply discounted price once certain conditions are met.

Staggered or classified board

Every director serves a three-year term and the terms are staggered so that only one -third of the directors are up for election each year.

White knights

When a hostile takeover appears to be inevitable, a target company will sometimes look for another, friendlier company to acquire it.

White squire

A large investor or firm agrees to purchase a substantial block of shares in the target with special voting rights. This action prevents a hostile raider from acquiring control of the target. The idea is that the white squire itself won't choose to exercise its control rights.

Golden parachute

An extremely lucrative severance package that's guaranteed to a firm's senior management in the event that the firm is taken over and the managers are let go.

The existence of golden parachutes lessens the likelihood of managerial entrenchment

Recapitalization

A company changes its capital structure to make itself less attractive as a target.

Other defensive strategies

- A corporation's charter can require a supermajority of votes to approve a merger

Regulatory approval

All mergers must be approved by regulators

28.6 WHO GETS THE VALUE ADDED FROM A TAKEOVER?

The premium the acquirer pays is approximately equal to the value it adds, which means the target shareholders ultimately capture the value added by the acquirer

When a bidder makes an offer for a firm, the target shareholders can benefit by keeping their shares and letting other shareholders sell at a low price. Because all shareholders have the incentive to keep their shares, no one will sell. This scenario is known as the **free rider problem**. To overcome this problem, bidders can acquire a toehold in the target, attempt a leveraged buyout, or, in the case when the acquirer is a corporation, offer a freezeout merger.

Toehold

An initial stake in the target

Freezeout merger

The laws on tender offers allow the acquiring company to freeze existing shareholders out of the gains from merging by forcing non-tendering shareholders to sell their shares for the tender offer price.

CHAPTER 29: CORPORATE GOVERNANCE

29.1 CORPORATE GOVERNANCE AND AGENCY COSTS

The system of controls, regulations, and incentives designed to prevent fraud

The different stakeholders in a firm all have their own interests. When those interests diverge, we may have agency conflicts.

The conflict of interest between managers and investors derives from the separation of ownership and control in a corporation. The separation of ownership and control is perhaps the most important reason for success of the corporate organizational form. Once control and ownership are separated, however, a conflict of interest arises. Agency conflicts are likely to arise any time the manager doesn't internalize the full cost of his or her actions.

The seriousness of this conflict of interest depends on how closely aligned the interests of the managers and shareholders are.

The role of corporate governance is to mitigate the conflict of interest that results from the separation of ownership and control without unduly burdening managers with the risk of the firm. The system attempts to align these interests by providing incentives for taking the right action and punishments for taking the wrong action. The incentives come from owning stock in the company and from compensation that's sensitive to performance.

29.2 MONITORING BY THE BOARD OF DIRECTORS

At first glance, one might think that there's a simple solution to the conflict of interest problem: monitor the firm's managers closely. The problem with this reasoning is that it ignores the cost of monitoring.

The shareholders as a group elect a board of directors to monitor managers. The board of directors hires the executive team, sets its compensation, approves major investments and acquisitions, and dismisses executives if necessary.

Three types of directors: inside, gray, and outside (or independent)

Inside directors

Employees, former employees, or family members of employees

Gray directors

People who aren't as directly connected to the firm as insiders are, but who have existing or potential business relationships with the firm.

Outside

All other directors. The most likely to make decisions solely in the interests of the shareholders.

Captured board

A board is said to be captured when its monitoring duties have been compromised by connections or perceived loyalties to management.

Board size and performance

Smaller boards are associated with greater firm value and performance.

Smaller groups make better decisions than larger groups.

Other monitors

The board is complemented by other monitors, both inside and outside the firm.

- Securities analysts produce independent valuations of the firms they cover so that they can make buy and sell recommendations to clients.
- Lenders also carefully monitor firms to which they're exposed to as creditors
- Employees of the firm are most likely to detect outright fraud because of their inside knowledge
- The SEC is charged with the task of protecting the investing public against fraud and stock manipulation.

29.3 COMPENSATION POLICIES

In the absence of monitoring, the other way the conflict of interest between managers and owners can be mitigated is by closely aligning their interests through the managers' compensation policy.

Stock and options

Bonuses based on earnings growth, stock and option grants.

Pay and performance sensitivity

Besides increasing managers' risk exposure, increasing the sensitivity of managerial pay and wealth to firm performance has some other negative effects.

Managers have an incentive to manipulate the release of financial forecasts so that bad news come out before options are granted (to drive the exercise price down) and good news come out after options are granted.

Backdating

The practice of choosing the grant date of a stock option retroactively, so that the date of the grant coincides with a date when the stock price was at its low for the quarter or for the year.

29.4 MANAGING AGENCY CONFLICT

Direct action by shareholders

- **Shareholder voice:** any shareholder can submit a resolution that's put to a vote at the annual meeting. A resolution could direct the board to take a specific action.
- **Shareholder approval:** shareholders must approve many major actions taken by the board.
- **Proxy contests:** the most extreme form of direct action that disgruntled shareholders can take is to hold a proxy contest and introduce a rival slate of directors for election to the board.

29.5 REGULATION

The Sarbanes-Oxley Act

The overall intent of the legislation was to improve the accuracy of information given to both boards and to shareholders.

SOX attempted to achieve this goal in three ways:

1. By overhauling incentives and independence in the auditing process
 2. By stiffening penalties for providing false information
 3. By forcing companies to validate their internal financial control processes
- Strict limits on the amount of non-audit fees that an accounting firm can earn from the same firm that it audits
 - Audit partners rotate every five years to limit the likelihood that auditing relationships become too cozy over long periods of time
 - Called on the SEC to force companies to have audit committees that are dominated by outside directors and required that at least one outside director have a financial background
 - Stiffened the criminal penalties for providing false information to shareholders
 - Requires senior management and the boards of public companies to be comfortable enough with the process through which funds are allocated and controlled, and outcomes monitored throughout the firm, to be willing to attest to their effectiveness and validity.

The Cadbury Commission

- Audit compensation committees should be made up entirely of independent directors, or at least, have a majority of them
- The CEO shouldn't be chairman of the board
- There should be a lead independent director with similar agenda-setting powers
- Auditors should be rotated
- There should be fuller disclosure of non-audit work

Unlike SOX, these recommendations weren't backed by the force of the law

Dodd-Frank Act

Was spurred by the 2008 financial crisis.

- **Independent compensation committee:** All US exchanged must require that the listed firms' compensation committees be composed of only independent board members, and have the authority to hire outside compensation consultants
- **Nominating directors:** large shareholders owning at least 3% of a company's stock for at least three years may nominate candidates for the board of directors, with the candidates listed on the firm's proxy statement alongside the nominees of management
- **Vote on Executive Pay and Golden Parachutes:** at least once every three years, firms must provide shareholders with a nonbinding vote on the compensation of the firm's CEO, CFO, and the three other most highly paid executives. While the vote is non-binding, companies must formally respond regarding how they have taken into account the results of the vote
- **Clawback provisions:** public companies must establish policies that allow firms to take back up to three years of any executive incentive compensation erroneously awarded in the event of an accounting restatement

- **Pay disclosure:** companies must disclose the ratio of CEO annual total compensation to that of the median employee. Companies are also required to disclose the relationship between executive compensation and the firm's financial performance. Finally, firms are required to disclose whether they permit employees and directors to hedge against decreases in the market value of the company's stock

Insider trading

Occurs when a person makes a trade based on privileged information.

Insiders: managers, directors, and anyone else who has access to material non-public information, including temporary insiders

Material information

Whether the information would have been a significant factor in an investor's decision about the value of the security

29.6 CORPORATE GOVERNANCE AROUND THE WORLD

Dual class shares

A scenario in which companies have more than one class of shares and one class has superior voting rights to the other class

Pyramid structure

In this structure, a family first created a company in which it owns more than 50% of the shares and therefore has a controlling interest. This company then owns a controlling interest, at least 50% of the shares in another company. So, the family controls both companies, but owns only 25% of the second company.

The farther you move down the pyramid the less ownership the family has, but it still remains in complete control for all the companies.

Tunneling

The family has an incentive to try to move profits up the pyramid, away from companies in which it has few cash flow rights and towards firms in which it has more cash flow rights

Stakeholder model

Give explicit consideration to other stakeholders, in particular, rank-and-file employees.

Cross-Holdings

Keiretsu: groups of firms connected through cross-holdings and a common relation to a bank
Chaebol: conglomerate groups of companies in widely diversified lines of business

29.7 THE TRADE-OFF OF CORPORATE GOVERNANCE

Corporate governance

A system of checks and balances that trades off costs and benefits.

This trade-off is very complicated, no one structure works for all firms.

Good governance is value enhancing and so, in principle, something investors in the firm should strive for

CHAPTER 30: RISK MANAGEMENT

30.1 INSURANCE

The most common method firms use to reduce risk.

Property insurance

Many firms use it to insure their assets against hazards such as fire, storm damage, vandalism, earthquakes, and other natural and environmental risks.

Other common types of insurance:

- **Business liability insurance** covers the costs that result if some aspect of the business causes harm to a third part or someone else's property
- **Business interruption insurance** protects the firm against the loss of earning if the business is interrupted due to fire, accident, or some other insured peril
- **Key personnel insurance** which compensates for the loss or unavoidable absence of crucial employees in the firm

When a firm buys insurance, it transfers the risk of the loss to an insurance company, which charges an upfront premium to take on that risk.

Insurance pricing in a perfect market

In a perfect market without other frictions, insurance companies should compete until they're just earning a fair return and the NPV from selling insurance is zero.

Actuarially fair insurance premium

$$\text{Insurance premium} = \frac{\text{Pr}(\text{Loss}) * E[\text{Payment} \in \text{the Event of Loss}]}{1 + r_L}$$

r_L is the appropriate cost of capital given the risk of the loss

$\text{Pr}(\text{Loss})$ the probability that the loss will occur

$E[]$ the expected payment if the loss occurs

The Value of Insurance

The value of insurance must come from reducing the cost of market imperfections on the firm

- Bankruptcy and financial distress costs
- Issuance costs
- Tax rate fluctuations
- Debt capacity
- Managerial incentives
- Risk assessment

The cost of insurance

When insurance premiums are actuarially fair, using insurance to manage the firm's risk can reduce costs and improve investment decisions.

But in reality, market imperfections exist that can raise the cost of insurance above the actuarially fair price and offset some of these benefits.

Three main frictions may arise between the firm and its insurer

1. Transferring the risk to an insurance company entails administrative and overhead costs.
2. Adverse selection.
3. Agency costs. Insurance reduces the firm's incentive to avoid risk.

Moral hazard

A change in behavior that results from the presence of insurance.

Example: after purchasing fire insurance, a firm may decide to cut costs by reducing expenditures on fire prevention

Insurance companies structure their policies in such a way as to reduce these costs.

Deductible

The initial amount of the loss that's not covered by insurance

Policy limits

Limit the amount of the loss that's covered regardless of the extent of the damage

These provisions mean that firms continue to bear some of the risk of the loss even after it's insured.

For insurance to be attractive, the benefit to the firm must exceed the additional premium charged by the insurer.